
An Energy-Based View on the Manifold Hypothesis

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 A commonly adopted assumption in modern machine learning is the *manifold*
2 *hypothesis*, which claims that although data is embedded in a high-dimensional
3 ambient space, the probability mass concentrates near a much lower-dimensional
4 geometric structure. However, the real-world latent variables that images are drawn
5 from are continuous, not discrete, and therefore cannot be completely captured by
6 a finite manifold. Moreover, the assumption does not properly capture the union
7 of manifolds, stratified spaces, and manifolds with singularities and boundaries.
8 Drawing upon recent advances in energy-based models, we provide an energy-
9 based perspective on the real-world latents, describing an Occam’s razor style
10 analog for the manifold hypothesis, which is more amenable to recent literature
11 in the field. We theoretically prove that our construction recovers the manifold
12 hypothesis for coarse functions, and validate it with numerical examples.

13 1 Introduction

14 If pictures are discretized objects in $\mathbb{R}^d$, where $d = n^2$ is the dimension of the image.

15 We consider raw objects $f : [0, 1]^2 \rightarrow \mathbb{R}^3$ corresponding to depixelated RGB pictures.

16 A commonly adopted assumption in modern machine learning is the *manifold hypothesis*, which
17 claims that although data is embedded in a high-dimensional ambient space, the probability mass
18 concentrates near a much lower-dimensional geometric structure.

19 **Assumption 1.1** (Manifold Hypothesis). Let $\chi \subset \mathbb{R}^D$ be the ambient space and let $\mathcal{D}$ be a probability
20 distribution on χ . There exists a d -dimensional manifold $\mathcal{M} \subset \chi$, with $d \ll D$, such that

$$\mathbb{P}_{x \sim \mathcal{D}}(\text{dist}(x, \mathcal{M}) \leq \varepsilon) \geq 1 - \delta,$$

21 for sufficiently small $\varepsilon, \delta > 0$.

22 In practice, data rarely concentrate on a single smooth manifold. Rather, they are better modeled as
23 lying on *unions of manifolds*, or more generally on *stratified spaces* composed of pieces of varying
24 intrinsic dimensions, potentially with singularities and boundaries.

25 Why small d is important?

- 26 • Generalization error depends on the intrinsic dimension d of the manifold.
- 27 • The number of samples n needed to learn the manifold increases with d .
- 28 • The complexity of the model increases with d .
- 29 • Can get a small embedding.

30 Some shortcomings of the manifold hypothesis are that

- 31 • Doesn’t properly capture union of manifolds.

- Doesn't properly capture stratified spaces.
- Doesn't properly capture manifolds with singularities and boundaries.
- What about infinite-dimensional data? Cause images come from infinite-dimensional continuous functions.

[Emile says: would be nice to elaborate on the above, and to point to specific instances where assuming it has some "catastrophic" consequences. that could go into a motivating examples section]

1.1 Related Works

Energy-based models Song and Kingma [2021], Du and Mordatch [2019], LeCun et al. [2006], Zhu et al. [2024], Du et al. [2021, 2020]

Compositional generation Su et al. [2024], Liu et al. [2021, 2022, 2023], Du and Kaelbling [2024], Bagautdinov et al. [2018]

Occam's razor Elmoznino et al. [2025], Rasmussen and Ghahramani [2000]

Example: Vision Data Each 2d image (or 3d scene or a point cloud) can be modeled $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ as a function from coordinates to RGB (or XYZ) values. That's how NeRFs work:

Neural Radiance Fields (NeRFs). Mildenhall et al. [2021], Pumarola et al. [2021], Xu et al. [2021], Wang et al. [2022], Rabby and Zhang [2024], Reiser et al. [2021], Müller et al. [2022], Tancik et al. [2021], Yu et al. [2021], Chen et al. [2022] [Emile says: this actually looks very relevant. would be worth doing a deeper dive into this] A *Neural Radiance Field* models a 3D scene as a continuous function

$$f_{\theta} : (\mathbf{x}, \mathbf{d}) \mapsto (\sigma, \mathbf{c}),$$

where

- $\mathbf{x} \in \mathbb{R}^3$ denotes a spatial location,
- $\mathbf{d} \in \mathbb{S}^2$ denotes a viewing direction,
- $\sigma \in \mathbb{R}_{\geq 0}$ is the volume density at $\mathbf{x}$,
- $\mathbf{c} \in \mathbb{R}^3$ is the emitted RGB color in direction $\mathbf{d}$.

The key idea is that the scene is represented neither as a mesh, nor as a voxel grid, nor as a point cloud, but as a *continuous radiance field* parameterized by a neural network.

In a similar way, each 2d image can be modeled as a function from coordinates to RGB values: $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ and the embedding of this function would be the weights of the corresponding ReLU network. The regular manifold hypothesis doesn't properly capture this structure that:

- In a union of manifolds
- Has some hierarchical structure.
- Has permutation-invariance with respect to weights in each layer.

Also, importantly, each image f is not a smooth function: its a combination of a "simple" in some sense partition of the image into smooth enough regions.

2 Problem Setting

Notation. We refer to a metric space (X, d) as a Polish space if it is complete (every Cauchy sequence in (X, d) converges to a point in X) and separable (there exists a countable dense subset $D \subset X$). For instance, $\mathbb{R}^n$ with the Euclidean metric, separable Banach spaces, Hilbert spaces, and function spaces such as $L^2(M)$ or $H^s(M)$ (when M is a compact manifold) are all Polish spaces. Moreover, if X is Polish, then $(X, \mathcal{B}(X))$ enjoys strong regularity properties that make it a canonical measurable structure in probability theory, where $\mathcal{B}(X)$ is the Borel σ -algebra on X .

Supervised Learning. From data $D = \{(x_i, y_i)\}_{i=1}^n$ we seek to learn a target function $f^* : \mathcal{X} \rightarrow \mathbb{R}$ such that $f^*(x_i) = y_i$ for all $i = 1, \dots, n$. Usually, we know the input space $\mathcal{X}$ via some base distribution ν_0 on $\mathcal{X}$, for instance, uniformly distribution over a compact set $\mathcal{X} \subset \mathbb{R}^d$. Moreover, we assume that smoother or more predictable the function f^* is, the better it generalizes to new data.

Definition 2.1 (Symmetric Dirichlet Space). Let $(X, \mathcal{F}, \mu)$ be a σ -finite measure space. A *Dirichlet form* on $L^2(\mu)$ is a pair $(\mathcal{E}, \mathcal{D}(\mathcal{E}))$ such that:

1. $\mathcal{D}(\mathcal{E}) \subset L^2(\mu)$ is a dense linear subspace;

2. $\mathcal{E} : \mathcal{D}(\mathcal{E}) \times \mathcal{D}(\mathcal{E}) \rightarrow \mathbb{R}$ is a symmetric bilinear form;

3. (Non-negativity)

$$\mathcal{E}(f, f) \geq 0 \quad \forall f \in \mathcal{D}(\mathcal{E});$$

4. (Closedness) $\mathcal{D}(\mathcal{E})$ is complete with respect to the norm

$$\|f\|_{\mathcal{E}}^2 := \|f\|_{L^2(\mu)}^2 + \mathcal{E}(f, f);$$

5. (Markov property) for every $f \in \mathcal{D}(\mathcal{E})$, the truncation $\tilde{f} := (0 \vee f) \wedge 1$ satisfies

$$\tilde{f} \in \mathcal{D}(\mathcal{E}) \quad \text{and} \quad \mathcal{E}(\tilde{f}, \tilde{f}) \leq \mathcal{E}(f, f).$$

The triple $(X, \mathcal{F}, \mu, \mathcal{E})$ is called a *Symmetric Dirichlet Space*.

The Dirichlet form encodes the geometry of the learning space via the energy functional: $\mathcal{E}(f, f)$ is the energy of the function f . Enforcing the Markov property allows us to define stochastic processes on the learning space: semi-group generated by $\mathcal{E}$ is a Markov semigroup.

The main idea of learning is that if $x_1 \approx x_2$ then $f(x_1) \approx f(x_2)$ for a function f . Dirichlet form enforces precisely this property. As it encodes the geometry of the learning space, it allows us to define a generalized Laplacian L on $L^2(\mu)$ and Sobolev classes of functions on X .

Definition 2.2 (Generalized Laplacian). Let $(X, \mathcal{F}, \mu, \mathcal{E})$ be a measure space endowed with a Dirichlet form. There exists a unique self-adjoint operator L on $L^2(\mu)$ such that

$$\mathcal{E}(f, g) = \langle f, Lg \rangle_{L^2(\mu)}, \quad \forall f \in \mathcal{D}(\mathcal{E}), g \in \mathcal{D}(L).$$

The operator L admits a spectral decomposition $L\phi_j = \lambda_j\phi_j$, with

$$0 = \lambda_0 \leq \lambda_1 \leq \lambda_2 \leq \dots, \quad \langle \phi_j, \phi_k \rangle_{L^2(\mu)} = \delta_{jk}.$$

The Markov property of the Dirichlet form $\mathcal{E}$ implies that e^{tL} is a Markov semigroup. Hence, our distribution μ is a stable state of the system: given a base distribution ν_0 on $\mathcal{X}$, the distribution μ is the limit of the Markov process $e^{tL}\nu_0$ as $t \rightarrow \infty$.

Definition 2.3 (Effective dimension). The *effective dimension* of the learning space $(X, \mathcal{F}, \mu, \mathcal{E})$ is the quantity $d_{\text{eff}} \in (0, \infty)$ such that the eigenvalues of the corresponding Laplacian operator $L : L^2(\mu) \rightarrow L^2(\mu)$ satisfy

$$\lambda_j \asymp j^{2/d_{\text{eff}}} \quad \text{as } j \rightarrow \infty,$$

or equivalently, the eigenvalue counting function satisfies

$$N(\lambda) := \#\{j : \lambda_j \leq \lambda\} \asymp \lambda^{d_{\text{eff}}/2} \quad \text{as } \lambda \rightarrow \infty.$$

Or, more generally

$$\text{tr } e^{tL} \sim t^{-d_{\text{eff}}/2} \quad \text{as } t \rightarrow 0^+$$

102 **Definition 2.4** (Spectral Sobolev class). For smoothness $s > 0$ and radius $R > 0$, the *spectral*
 103 *Sobolev class* of order s is

$$\mathcal{H}_\mu^s(R) := \left\{ f \in L^2(\mu) : \|f\|_{\mathcal{H}^s}^2 := \sum_{j \geq 1} \lambda_j^s \widehat{f}_j^2 \leq R^2 \right\}, \quad \widehat{f}_j := \langle f, \phi_j \rangle_{L^2(\mu)}.$$

104 The condition $f \in \mathcal{H}_\mu^s(R)$ requires the spectral coefficients to decay as $|\widehat{f}_j| \leq R \lambda_j^{-s/2} \asymp R j^{-s/d_{\text{eff}}}$:
 105 higher smoothness s enforces faster decay and hence greater regularity with respect to the geometry
 106 encoded by $\mathcal{E}$.

107 **Definition 2.5** (Heat semigroup Besov space). Let $P_t = e^{-tL}$ be the Markov semigroup generated
 108 by L . For $s > 0$ and $1 \leq p, q \leq \infty$, define

$$\|f\|_{B_{p,q}^s(L)} = \|f\|_{L^p(\mu)} + \left(\int_0^1 \left(t^{-s/2} \|(I - P_t)f\|_{L^p(\mu)} \right)^q \frac{dt}{t} \right)^{1/q}.$$

109 Then $B_{p,q}^s(L)$ consists of all f for which this norm is finite.

110 **Definition 2.6** (Spectral wavelets associated to L). Let $\varphi_0 \in C_c^\infty([0, \infty))$ satisfy $\varphi_0(\lambda) = 1$ for
 111 $\lambda \in [0, 1]$ and $\text{supp } \varphi_0 \subset [0, 2]$. Define the dyadic wavelet kernel

$$\psi(\lambda) = \varphi_0(\lambda) - \varphi_0(2\lambda), \quad \psi_j(\lambda) = \psi(2^{-j}\lambda), \quad j \geq 1,$$

112 so that $\{\varphi_0, \psi_j\}_{j \geq 1}$ forms a partition of unity:

$$\varphi_0(\lambda) + \sum_{j=1}^{\infty} \psi_j(\lambda) = 1 \quad \text{for all } \lambda \geq 0.$$

113 The *wavelet blocks* are the operators

$$\Psi_0 = \varphi_0(L), \quad \Psi_j = \psi_j(L) = \psi(2^{-j}L), \quad j \geq 1,$$

114 defined via the spectral calculus of L , yielding the Littlewood–Paley decomposition

$$f = \Psi_0 f + \sum_{j=1}^{\infty} \Psi_j f.$$

115 If L has discrete spectrum $0 \leq \lambda_1 \leq \lambda_2 \leq \dots$ with eigenfunctions $\{e_k\}$, then

$$\Psi_j f = \sum_k \psi_j(\lambda_k) \langle f, e_k \rangle_{L^2(\mu)} e_k.$$

116 3 Effective Dimension Hypothesis

117 Let us formulate the extension of the manifold hypothesis to this case. [\[Emile says: more generally,](#)
 118 [need to define energy functions, why they're relevant to this problem, and what characteristics do we](#)
 119 [want our manifold hypothesis to have?\]](#)

120 **Assumption 3.1** (Effective Dimension Hypothesis). All the observed data x_i come from a learning
 121 space $(X, \mathcal{F}, \mu, \mathcal{E})$ with small effective dimension even if the ambient space $(X, \mathcal{F}, \nu_0, \mathcal{E})$ has large
 122 effective dimension where ν_0 is the base distribution on $\mathcal{X}$:

$$d_{\text{eff}}(X, \mathcal{F}, \mu, \mathcal{E}) \ll d_{\text{eff}}(X, \mathcal{F}, \nu_0, \mathcal{E})$$

123 This assumption ensures two things: existence of the efficient coordinate system in the space $\mathcal{X}$ and
 124 efficient learning rate for functions in Besov spaces. Moreover, this assumption of the underlying
 125 learning space is strictly stronger than the manifold hypothesis and unifies multiple learning problems,
 126 e.g. classical kernel methods, flow matching/diffusion models and transformers.

127 3.1 Generalization Error

128 **Proposition 3.2** (Bias-variance tradeoff). *Let $f^* \in \mathcal{H}_\mu^s(R)$ and suppose we observe $y_i = f^*(x_i) + \varepsilon_i$,
129 $i = 1, \dots, n$, with $x_i \sim \mu$ i.i.d. and ε_i i.i.d. sub-Gaussian with parameter σ^2 . Consider the truncated
130 spectral estimator*

$$\widehat{f}_k(x) = \sum_{j=1}^k \widehat{f}_j^{(n)} \phi_j(x), \quad \widehat{f}_j^{(n)} = \frac{1}{n} \sum_{i=1}^n y_i \phi_j(x_i).$$

131 *Then the expected risk decomposes as*

$$\mathbb{E}[\|\widehat{f}_k - f^*\|_{L^2(\mu)}^2] = \underbrace{\sum_{j>k} (\widehat{f}_j^*)^2}_{\text{bias}^2(k)} + \underbrace{\frac{\sigma^2 k}{n}}_{\text{variance}(k)}.$$

132 *Under Effective Dimension Hypothesis (Assumption 3.1) ($\lambda_j \asymp j^{2/d_{\text{eff}}}$) and the Sobolev constraint
133 $\sum_j \lambda_j^s (\widehat{f}_j^*)^2 \leq R^2$:*

134 **Bias bound.** *The tail energy is controlled by the smoothness and dimension:*

$$\text{bias}^2(k) = \sum_{j>k} (\widehat{f}_j^*)^2 \leq R^2 \lambda_{k+1}^{-s} \asymp R^2 k^{-2s/d_{\text{eff}}}.$$

135 **Variance bound.**

$$\text{variance}(k) = \frac{\sigma^2 k}{n}.$$

136 **Optimal truncation.** *Balancing bias and variance:*

$$k^{-2s/d_{\text{eff}}} \asymp \frac{k}{n} \implies k^* \asymp n^{d_{\text{eff}}/(2s+d_{\text{eff}})}.$$

137 **Minimax rate.** *Substituting k^* into either term:*

$$\boxed{\mathbb{E}[\|\widehat{f}_{k^*} - f^*\|_{L^2(\mu)}^2] \lesssim n^{-2s/(2s+d_{\text{eff}})}}.$$

138 *This rate is minimax optimal over $\mathcal{H}_\mu^s(R)$.*

139 **Remark 3.3** (Interpretation). The three quantities s , d_{eff} , and n interact as follows:

- 140 • **Smoothness s :** higher s means faster eigenvalue decay of the coefficients $\widehat{f}_j^*$, so the bias
141 decreases faster with k and fewer modes suffice.
- 142 • **Effective dimension d_{eff} :** larger d_{eff} means eigenvalues $\lambda_j \asymp j^{2/d_{\text{eff}}}$ grow more slowly, so
143 more modes carry significant energy and the variance cost of including them is higher. This
144 is the curse of dimensionality in d_{eff} , not in the ambient dimension.
- 145 • **Optimal number of modes k^* :** this is the resolution at which you can reliably estimate the
146 function given n samples. Modes $j \leq k^*$ are estimated; modes $j > k^*$ are discarded as
147 noise.

148 **Theorem 3.4** (Minimax rate over Besov balls ?). *[TODO: Replace $\mathbb{R}^d$ by $\mathcal{X}$] Let $f \in \mathcal{B}_{p,q}^s(R) \subset$
149 $L^r(\mathbb{R}^d)$ with $s > d(1/p - 1/r)_+$. Then the minimax L^r estimation rate from n noisy samples is*

$$\inf_{\widehat{f}_n} \sup_{f \in \mathcal{B}_{p,q}^s(R)} \mathbb{E}[\|\widehat{f}_n - f\|_{L^r}] \asymp \begin{cases} n^{-\frac{s}{2s+d}}, & \text{if } p \geq r \quad (\text{dense zone}), \\ \left(\frac{\log n}{n}\right)^{\frac{s-d(1/p-1/r)}{2(s-d(1/p-1/r))+d}}, & \text{if } p < r \quad (\text{sparse zone}). \end{cases}$$

150 *The dense rate is achieved by linear methods (projection, kernel smoothing). The sparse rate requires
151 nonlinear wavelet thresholding; linear estimators are suboptimal by a polynomial factor in $\log n$.*

152 3.2 Compression

153 **Theorem 3.5.** *Lower bound on the number of bits to represent a function f up to the accuracy ϵ is*
 154 *given by*

$$\text{bits}(f, \epsilon) \geq \dots$$

155 4 Set Up

156 Common belief about the data but make it abstract:

- 157 • Observed data $x_1, \dots, x_n$ comes from some space X (e.g. $\mathbb{R}^D$)
- 158 • We know some "sparse" distribution over X : ν_0 (e.g. Gaussian, Uniform)
- 159 • However, we believe that the data comes from some "compact" or "concentrated" distribution
 160 μ over X (e.g. d -dim manifold $M \subset \mathbb{R}^D$)

161 If that is the case, we have better hope to learn functions $f : X \rightarrow \mathbb{R}$: more concentration – faster
 162 rates.

163 First, define the data space and equip it with the differentiation property.

- 164 • Let X be a set
- 165 • an algebra of functions $f : X \rightarrow \mathbb{R}, \mathcal{A}$
- 166 • $D : \mathcal{A} \rightarrow \mathcal{M}$ a derivation into some module $\mathcal{M}$ equipped with a pointwise inner product
 167 $\langle \cdot, \cdot \rangle_x$
- 168 • Leibniz rule: $D(fg) = fDg + gDf$.
- 169 • Hence, the carre du champ operator is $\Gamma(f, g)(x) = \langle Df, Dg \rangle_x$. It generalizes square-
 170 gradient concept.

171 Now let us add a measure ν over X . That defines:

- 172 • Dirichle form $\mathcal{E}_\nu(f, g) = \int_X \Gamma(f, g) d\nu(x)$ ([TODO: maybe add some assumptions on this])
- 173 • Self-adjoint Laplacian L_ν such that $\mathcal{E}_\nu(f, f) = \langle f, L_\nu f \rangle_\nu$
- 174 • Markov semi-group: e^{-tL_ν} , it defines a diffusion that converges to ν .
- The iterated carré du champ (or Γ_2^ν operator) is:

$$\Gamma_2^\nu(f, g) = \frac{1}{2} (L_\nu \Gamma(f, g) - \Gamma(f, L_\nu g) - \Gamma(g, L_\nu f))$$

and

$$\Gamma_2^\nu(f) := \Gamma_2^\nu(f, f) = \frac{1}{2} (L_\nu |Df|^2 - 2\langle Df, D(L_\nu f) \rangle)$$

175 Alright, what defines the "concentration" of a measure ν ? What structural assumption would give us
 176 hope to learn functions on it?

177 5 Key Assumptions

Definition 5.1 (Curvature–Dimension Condition). For $\rho \in \mathbb{R}, N \in (0, \infty)$, ν satisfies $CD(\rho, \Gamma)$ if for all $f \in \mathcal{A}$

$$\Gamma_2^\nu(f, f) \geq \rho \Gamma(f, f) + \frac{1}{N} (L_\nu f)^2$$

178 Intuitively: $\rho \sim$ curvature parameter, $N \sim$ dimension

179 5.1 Sub-Exponential Concentration

Definition 5.2 (Log-Sobolev Inequality). Given μ for there is a constant C_{LS} such that the following holds for any $f : X \rightarrow \mathbb{R}$

$$\text{Ent}_\mu(f^2) \leq 2C_{LS} \int \|Df\|^2 d\mu$$

180 where $\text{Ent}_\mu(g) = \int g \log g d\mu - \int g d\mu \cdot \log \int g d\mu$ for $g \geq 0$.

181 **Lemma 5.3.** Assuming $\Gamma_2^\mu(f, f) \geq \rho \Gamma(f, f)$ for all $f \in \mathcal{A}$ then $C_{LS} \leq \frac{1}{\rho}$

182 Consequences:

Theorem 5.4 (Hypercontractivity, Gross 1975). *The LSI is equivalent to hypercontractivity of the semigroup: for $1 < p < q < \infty$,*

$$\|P_t f\|_{L^q(\mu)} \leq \|f\|_{L^p(\mu)} \quad \text{when } t \geq \frac{C_{LS}}{2} \log \frac{q-1}{p-1}$$

Theorem 5.5 (Sub-Gaussian concentration). *For f with $\Gamma(f, f) \leq 1, \mu$ -a.e., the Herbst argument gives*

$$\mu(f - \mathbb{E}_\mu f \geq t) \leq \exp\left(-\frac{t^2}{2C_{LS}}\right)$$

183 The smaller C_{LS} – the better.

184 5.2 Effective Dimension

185 **Definition 5.6** (Effective dimension). The *effective dimension* of $(X, \mathcal{A}, \Gamma, \mu)$ is the quantity $d_{\text{eff}} \in$
186 $(0, \infty)$ such that the eigenvalues of the corresponding Laplacian operator $L : L^2(\mu) \rightarrow L^2(\mu)$ satisfy

$$\lambda_j \asymp j^{2/d_{\text{eff}}} \quad \text{as } j \rightarrow \infty,$$

187 or equivalently, the eigenvalue counting function satisfies

$$N(\lambda) := \#\{j : \lambda_j \leq \lambda\} \asymp \lambda^{d_{\text{eff}}/2} \quad \text{as } \lambda \rightarrow \infty.$$

188 Or, more generally

$$\text{tr } e^{tL} \sim t^{-d_{\text{eff}}/2} \quad \text{as } t \rightarrow 0^+$$

189 The smaller d_{eff} – the better.

190 **Corollary 5.7.** If ν satisfies $CD(\rho, \Gamma)$ then $d_{\text{eff}} \leq N$

191 6 Supervised Learning: Learning a Function From Data

192 Adding a prior $p(f) \propto e^{-E(f)}$ over functions $f : X \rightarrow \mathbb{R}$ the supervised learning problem is
193 equivalent to the following optimization problem:

$$\hat{f}_n = \min_f \sum_{i=1}^n \|f(x_i) - y_i\|^2 + E(f)$$

194 Note, that $E(f)$ defines sublevel sets $\mathcal{F}_r = \{f : E(f) \leq r\}$. The generalization error of $\hat{f}_n$
195 decomposes as:

$$\mathbb{E} \left\| f^* - \hat{f}_n \right\|_{L^2}^2 \leq \underbrace{\inf_{f: E(f) \leq r} \|f - f^*\|^2}_{\text{bias: how well } \mathcal{F}_r \text{ approximates } f^*} + \underbrace{\varepsilon_n^2(r)}_{\text{variance: estimation error within } \mathcal{F}_r}$$

Definition 6.1 (Sublevel set). Given a convex functional $E : \mathcal{A} \rightarrow \mathbb{R} \cup \{+\infty\}$, a radius $r > 0$, and a metric d on $\mathcal{A}$, the sublevel set is:

$$\mathcal{B}_r(E) = \{f \in \mathcal{A} : E(f) \leq r\}$$

196 **Definition 6.2** (Covering number). The ϵ -covering number is:

$$N(\epsilon, \mathcal{B}_r(E), d) = \min \left\{ m \in \mathbb{N} : \exists f_1, \dots, f_m \in \mathcal{A} \text{ s.t. } \mathcal{B}_r(E) \subseteq \bigcup_{i=1}^m \{g : d(g, f_i) \leq \epsilon\} \right\}$$

197 The natural metric in our setting is $d(f, g) = \|f - g\|_{L^2(\mu)} = (\int_X (f - g)^2 d\mu)^{1/2}$

Definition 6.3 (Metric entropy).

$$\mathcal{H}(\epsilon, \mathcal{B}_r(E), d) = \log N(\epsilon, \mathcal{B}_r(E), d)$$

198 The smaller the metric entropy – the less samples we need to learn a function from data.

199 Lets consider 3 simple enough cases of what $E(f)$ can be:

200 1. Given convex $g : \mathbb{R} \rightarrow \mathbb{R}$ define $E(f) = \int_X g(f(x)) d\mu(x)$

201 2. $E(f) = \mathcal{E}_\mu(f, f)$

202 3. Given $\phi : \mathbb{R} \rightarrow \mathbb{R}$ define $E(f) = \langle f, \phi(L_\mu)f \rangle$

203 **Proposition 6.4.** For $E(f) = \int_X g(f(x)) d\mu(x)$ The covering number of $\mathcal{B}_r(E)$ is at most
 204 $N(r, E) = \frac{1}{r} \int_0^r \mathcal{H}(\epsilon, \mathcal{B}_r(E), d) d\epsilon$.

205 As we introduced Definition 5.1 that captures the "compactness" of the measure, we can say now that
 206 $\mu \ll \nu$ if

$$\begin{aligned} \mu &\in CD(\rho_\mu, N_\mu), \nu \in CD(\rho_\nu, N_\nu) \\ &\text{and} \\ \rho_\mu &> \rho_\nu, N_\mu < N_\nu \end{aligned}$$

207 The following theorem captures what density $\frac{d\mu}{d\nu}$ "simplifies" the measure.

208 **Theorem 6.5** (Perturbation of $CD(\rho, N)$). *[TODO: double check, add a proof, better notation]*

209 Let $(X, \mathcal{A}, \Gamma, \mu_0)$ be a Markov diffusion structure with generator L_0 satisfying $CD(\rho, N)$:

$$\Gamma_2^{(0)}(f, f) \geq \rho \Gamma(f, f) + \frac{1}{N} (L_0 f)^2 \quad \text{for all } f \in \mathcal{A}$$

210 Let $U \in \mathcal{A}$ satisfy:

- 211 • Abstract κ -convexity: $\Gamma(f, \Gamma(U, f)) - \frac{1}{2} \Gamma(U, \Gamma(f, f)) \geq \kappa \Gamma(f, f)$ for all f
- 212 • Bounded abstract gradient: $\Gamma(U, U) \leq M$ μ_0 -a.e.

213 Let $d\mu = Z^{-1} e^{-U} d\mu_0$ with $L_U f = L_0 f - \Gamma(U, f)$. Then for any $\varepsilon \in (0, 1)$, (L_U, Γ, μ) satisfies
 214 $CD(\rho', N')$ with

$$\rho' = \rho + \kappa - \frac{(1/\varepsilon - 1)M}{N}, \quad N' = \frac{N}{1 - \varepsilon}$$

215 In particular, $C_{LS}(\mu) \leq \frac{1}{\rho'}$ whenever $\rho' > 0$.

216 7 Diffusion Models

217 *[TODO: Add noise schedule maybe.]*

218 **Forward process:** $\mu \rightarrow \mu_0$ Density $p_t = \mu_t/\mu_0$ evolves according to the diffusion equation:

$$\frac{\partial p_t}{\partial t} = L_{\mu_0} p_t \implies p_t = e^{tL_{\mu_0}} \mu_0$$

Backward process: $\mu_0 \rightarrow \mu$ Density p_t evolves according to the diffusion+drift equation:

$$\frac{\partial p_t}{\partial t} = L_{\mu_0} p_t - \nabla \cdot (p_t \nabla \log p_t)$$

219 To be able to reverse it back, we need to learn $\nabla \log p_t$. Let us look at the very first δ step of the
 220 forward process (corresponding to the very last δ step of the backward process) and learn a score
 221 function for that step $s(\cdot) = \nabla \log p_T = \nabla \log \frac{d\mu}{d\nu}$. The minimization problem is:

$$\min_s \underbrace{\mathbb{E}_{p_T} \left[\|s(\cdot) - \nabla \log p_T\|^2 \right]}_{\text{data term}} + \underbrace{R(s)}_{\text{regularizer term}} \quad (1)$$

222 While the data term is straightforward, the regularizer term is to be chosen. Assuming $d\mu =$
 223 $Z^{-1} e^{-U} d\mu_0 \implies \nabla \log p_T = -DU$. Our prior (hope) for U is that it concentrates the measure better,
 224 specifically, that by Definition 5.1 $U \in \mathcal{A}$ satisfies:

- 225 • κ -convexity: $\Gamma(f, \Gamma(U, f)) - \frac{1}{2} \Gamma(U, \Gamma(f, f)) \geq \kappa \Gamma(f, f)$ for all f
- 226 • Bounded gradient: $\Gamma(U, U) \leq M$ μ_0 -a.e.

227 Define the abstract Hessian as a map from 1-forms to bilinear forms on $D(\mathcal{A})$:

$$H_v(f) := \langle Df, D\langle v, Df \rangle \rangle - \frac{1}{2} \langle v, D\langle Df, Df \rangle \rangle$$

228 This is linear in v : $H_{\alpha v_1 + \beta v_2}(f) = \alpha H_{v_1}(f) + \beta H_{v_2}(f)$. Note, that the term $\langle v, v \rangle \leq M$ is
 229 quadratic in v .

Hence, the following set is convex:

$$\mathcal{C}_{\kappa, M} = \{v \in D(\mathcal{A}) : H_v(f) \geq \kappa \Gamma(f, f) \text{ and } \langle v, v \rangle \leq M \quad \forall f \in \mathcal{A}\}$$

230 First let us compute its covering number.

231 **Lemma 7.1.** *The covering number of $\mathcal{C}_{\kappa, M}$ is at most $N(M, \kappa) = \frac{M}{\kappa}$.*

232 **Lemma 7.2** (Very rough sketch). *Define $W_{\kappa, M}$ as [TODO: a fixed set of very cool points from $\mathcal{C}_{\kappa, M}$].*
 233 *Then for every $v \in \mathcal{C}_{\kappa, M}$ the following problem has a unique solution:*

$$\min_{\omega \in \mathfrak{M}_+(W_{\kappa, M})} \|\omega\|_{TV} \text{ such that } \int_{w \in W_{\kappa, M}} w d\omega(w) = f$$

234 Hence, if we know the set $W_{\kappa, M}$ then Equation (1) can be rewritten as:

$$\min_{\omega \in \mathfrak{M}_+(W)} \mathbb{E}_{p_T} \left[\left\| \int_{w \in W_{\kappa, M}} w d\omega(w) - \nabla \log p_T \right\|^2 \right] + \|\omega\|_{TV}$$

235 7.1 References

236 <https://djalil.chafai.net/blog/2023/01/12/log-sobolev-and-bakry-emery/>

8 Energy-Based Model Perspective on Effective Dimension

need framework that shows that

- pure noise

- fractals

- matryoshka

should be hard to learn

8.1 ASTs

Definition 8.1 (AND–OR Abstract Syntax Tree). Let $\mathcal{X}$ be an input space and $\phi : \mathcal{X} \rightarrow \mathbb{R}^d$ an embedding. An *AND–OR abstract syntax tree* (AST) is defined recursively as:

- **Atom (Leaf):**

$$T(x) = g(x), \quad g \in \mathcal{G},$$

where $\mathcal{G}$ is a class of simple local models (e.g. linear or affine functions).

- **AND node:**

$$T(x) = \mathbb{I}[C_1(x) \wedge \cdots \wedge C_k(x)] \cdot T'(x),$$

where each $C_i(x)$ is a constraint and T' is a subtree.

- **OR node:**

$$T(x) = \sum_{j=1}^m \mathbb{I}[C_j(x)] \cdot T_j(x),$$

where $\{C_j\}_{j=1}^m$ form a partition of $\mathcal{X}$ and T_j are subtrees.

The constraints $C(x)$ are typically hyperplane tests of the form

$$C(x) := \langle w, \phi(x) \rangle \geq 0.$$

Semantics.

- AND nodes represent conjunctions of constraints and correspond to intersections of half-spaces or cross-products of independent factors.
- OR nodes represent partitions of the input space into disjoint regions.
- Each root-to-leaf path defines a region

$$\mathcal{X}_\ell = \bigcap_{C \in \text{path}(\ell)} C,$$

on which the function reduces to a simple local model.

Example: 1D Piecewise Linear Function Let

$$f(x) = \begin{cases} a_1x + b_1, & x \leq t, \\ a_2x + b_2, & x > t. \end{cases}$$

This function admits the AND–OR representation

$$f(x) = \mathbb{I}[x \leq t] (a_1x + b_1) + \mathbb{I}[x > t] (a_2x + b_2).$$

Equivalently, as an AST:

$$\text{OR}\left(\text{AND}(x \leq t, \text{LINEAR}(a_1, b_1)), \text{AND}(x > t, \text{LINEAR}(a_2, b_2))\right).$$

262 **Example: Hierarchical Partition** Let $t_1 < t_2$ and

$$f(x) = \begin{cases} a_1x + b_1, & x \leq t_1, \\ a_2x + b_2, & t_1 < x \leq t_2, \\ a_3x + b_3, & x > t_2. \end{cases}$$

263 Then f can be written as

$$f(x) = \mathbb{I}[x \leq t_2] \left(\mathbb{I}[x \leq t_1](a_1x + b_1) + \mathbb{I}[t_1 < x \leq t_2](a_2x + b_2) \right) + \mathbb{I}[x > t_2](a_3x + b_3).$$

264 This corresponds to a depth-2 AND-OR tree.

265 **Example: ReLU as AND-OR** For $w \in \mathbb{R}^d$,

$$\text{ReLU}(\langle w, x \rangle) = \mathbb{I}[\langle w, x \rangle \geq 0] \langle w, x \rangle + \mathbb{I}[\langle w, x \rangle < 0] \cdot 0.$$

266 Thus a ReLU unit is an OR node with two AND branches. A finite ReLU network corresponds to a
267 (possibly large) AND-OR tree, whose leaves are linear functions and whose OR structure encodes
268 the activation pattern partition.

269

270 [Emile says: What kind of non-trivial theoretical results to validate this construction would be the
271 most insightful? One thing is that for coarse/non-granular functions, it should sort of recover the
272 original manifold hypothesis? What are the theoretical results pertaining to manifold hypothesis in
273 the other papers we've been reading? Can we establish any analogs?]

274 [Emile says: What is the simplest possible way to empirically validate anything that we've constructed
275 (after constructing it)?]

276 **Theorem 8.2.** *equivalency between effective dimension and energy?*

277 9 Examples

278 10 Conclusion

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